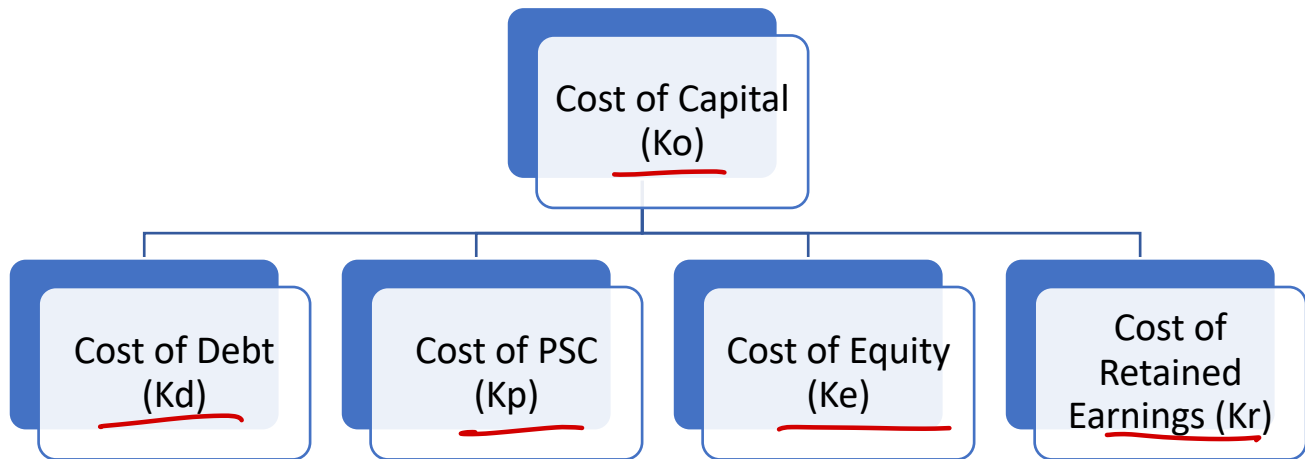


COST OF CAPITAL

1. Cost of Capital

- It is the weighted average of cost of various sources from which capital is raised.
- It is the minimum return to be earned by the company to meet the expectations of the capital providers.



2. Basic Terms

- ✓ **I** = Interest rate
 - ✓ **T** = Tax Rate
 - ✓ **RV** = Redeemable value
 - ✓ **N** = No. of years to maturity
 - ✓ **PD** = Preference dividend
 - ✓ **NP = P0** = Net Proceeds / Issue Price / Market Value / Face value
- If issue price not given then assume that issue price = market price
- Flotation cost to be considered on issue price if question is silent.
- Redeemable value to be taken at par value if question is silent.

3. Cost of Irredeemable Debt

$$K_d = \frac{I(1-t)}{NP} \times 100 \quad \checkmark$$

4. Cost of Redeemable Debt – Approximation Method

$$K_d = \frac{I(1-t) + \left(\frac{RV - NP}{n} \right)}{\left(\frac{NP + RV}{2} \right)} \times 100 \quad \checkmark$$

5. Cost of Redeemable Debt – YTM Method**If rates of PVF not given in question**

Find Kd using approximation method say x.y%

Find NPV at x% and (x + 1)%

If rates of PVF given in question

Use the rates given in question so that one NPV is +ve and other is -ve.

$$\text{NPV} = \text{PVC I} - \text{PVCO}$$

$$= [\text{Int.}(1 - t) \times \text{PVAF}_{(r,n)}] + [\text{RV} \times \text{PVF}_{(r,n)}] - \text{Cost today}$$

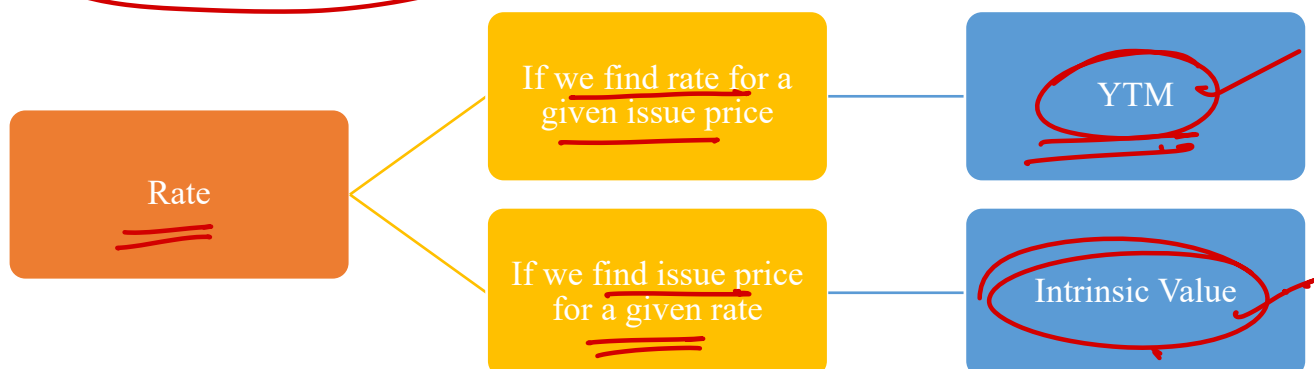
$$\text{Kd} = \text{IRR} = \text{Lower rate} + \frac{\text{Lower rate NPV}}{(\text{Lower rate NPV} - \text{Higher Rate NPV})} \times (\text{High Rate} - \text{low Rate})$$

6. Cost of Redeemable Debt in instalment

Calculate cash flows of each year

$$\text{Cash flow} = [\text{Interest} \times (1 - t)] + \text{Amortized maturity amount p.a.}$$

$$\text{Kd} = \text{IRR} = \text{Lower rate} + \frac{\text{Lower rate NPV}}{(\text{Lower rate NPV} - \text{Higher Rate NPV})} \times (\text{High Rate} - \text{low Rate})$$

7. YTM vs Intrinsic Value

Intrinsic Value (IV) = Present value of all future cash inflows

IV of bond/debenture = PV of interest + PV of redemption value

8. Decision on basis of Intrinsic value (IV)

(A) If IV > Current price	Undervalued	Recommend to Buy
(B) If IV < Current price	Overvalued	Recommend not to Buy

$$PV = \begin{cases} \text{Cash} = ₹ 100 \\ \text{Equity} = 5 \text{ Sh.} \times 20 \times (1 + 0.04)^5 = 121.65 \end{cases}$$

9. Convertible Debentures

Redeemable value = Higher of either cash or equity value

Value of one equity share = $P_0 \times (1 + g)^n$

$$K_d = \frac{10(1-0.25) + \frac{(121.65-100)}{5}}{\frac{(100+121.65)}{2}} =$$

Question – 1

TT Ltd. issued 20,000, 10% convertible debentures of ₹ 100 each with a maturity period of 5 years. At maturity the debentures holders will have the option to convert debentures into equity shares of the company in ratio of 1:5 (5 shares for each debentures). The current market price of the equity share is ₹ 20 each and historically the growth rate of the share is 4% per annum. Assuming tax rate is 25%. Compute the cost of 10% convertible debenture using Approximation Method and Internal Rate of Return Method.

PV Factor are as under:

Year	1	2	3	4	5
PV Factor @10%	0.909	0.826	0.751	0.683	0.621
PV Factor @15%	0.870	0.756	0.658	0.572	0.497

Solution

Value of equity shares after 5 years = $20 \times (1 + 0.04)^5 = ₹ 24.33$

Redemption value of debenture will be higher of:

a) Cash value of debenture = ₹ 100

b) Value of equity shares = $5 \times 24.33 = ₹ 121.65$

∴ Higher redemption value of the above two = ₹ 121.65

Approximation Method:

$$\text{Cost of Debentures (Kd)} = \frac{I(1-t) + \{(RV - NP) \div n\}}{\{(NP + RV) \div 2\}} = \frac{10(1-0.25) + \{(121.65 - 100) \div 5\}}{\{(100 + 121.65) \div 2\}} = \frac{11.83}{110.825} = 10.67\%$$

Internal Rate of Return Method:

Year	Cash Flows	Present Value		Present Value	
		PVF @ 10%	PV	PVF @ 15%	PV
0	→ 100	✓ 1.000	(100)	- 1.000	(100)
1-5	→ 7.5	✓ 3.790	28.43	- 3.353	25.15
5	→ 121.65	✓ 0.621	75.54	0.497	60.46
NPV			3.97		(14.39)

$$\begin{aligned} \text{Cost of Debentures (Kd)} &= L + \left[\frac{NPV_L}{NPV_L - NPV_H} \right] (H - L) \\ &= 10 + \left[\frac{3.97}{3.97 - (-14.39)} \right] (15 - 10) = 11.08\% \end{aligned}$$

Question – 2

SK Ltd. issued 12% Bonds of face value ₹ 2,000 each, which are redeemable after 5 years. Tax rate is 30% and the bonds are amortized equally over the life of bonds. Compute the value of the bond if the investor expects a minimum return of 8% from the bonds. $\rightarrow P.a. = \frac{2000}{5} = 400 \rightarrow IV$
 $\rightarrow \text{Given Return}$

Solution

$(100 - 30\%) = 70\%$

Year	Principal Outstanding	Principal Repayment	Interest Payment Net of Tax	Total Cash Flows	PVF @ 8%	PV
1	2,000	-400	$2,000 \times 12\% \times 70\% = 168$	568	0.926	525.97
2	$2000 - 400 = 1,600$	400	$1,600 \times 12\% \times 70\% = 134.4$	534.40	0.857	457.98
3	$1600 - 400 = 1,200$	400	$1,200 \times 12\% \times 70\% = 100.80$	500.80	0.794	397.64
4	800	400	$800 \times 12\% \times 70\% = 67.20$	467.20	0.735	343.39
5	400	400	$400 \times 12\% \times 70\% = 33.60$	433.60	0.681	295.28
					$IV =$	2020.26

Value of the bond = ₹2020.26

$$\frac{14.58}{100} = \frac{18(1-0.30) + \left(\frac{100-NP}{8}\right)}{\left(\frac{NP+100}{2}\right)} \Rightarrow NP = ?$$

Question – 3

A company issues 20,000, 18% Debentures of ₹100 each. The debentures are redeemable after a period of 8 years. The cost of debentures using approximation method is 14.58%. The corporate tax rate is 30%. You are required to calculate:

- Issue price of the Debentures, if redeemable at par.
- Revised Cost of Debentures using approximation method, if redeemable at 10% Premium after 8 years.

$$RV = 100 + 10\% = 110$$

Solution

- Cost of redeemable debenture (Kd) = $\frac{I(1-t) + [(RV-NP) \div n]}{(RV+NP) \div 2}$

$$0.1458 = \frac{18(1-0.3) + [(100-NP) \div 8]}{(100+NP) \div 2}$$

$$0.1458 \times \left(\frac{100+NP}{2}\right) = 12.6 + \left(\frac{100-NP}{8}\right)$$

$$7.29 + (0.0729)(NP) = 12.6 + 12.5 - (0.125)(NP)$$

$$0.1979(NP) = 17.81$$

$$NP = 90$$

- Revised cost of redeemable debenture (Kd) = $\frac{I(1-t) + [(RV-NP) \div n]}{(RV+NP) \div 2}$

$$= \frac{18(1-0.3) + [(110-90) \div 8]}{(110+90) \div 2} = 15.1\%$$

10. Cost of Irredeemable Preference Shares

$$K_p = \frac{PD}{NP} \times 100$$

11. Cost of Redeemable Preference Shares – Approximation Method

$$K_p = \frac{PD + \left(\frac{RV - NP}{n} \right)}{\left(\frac{NP + RV}{2} \right)} \times 100$$

12. Cost of Redeemable Preference Shares – YTM Method

$$K_p = IRR = \text{Lower rate} + \frac{\text{Lower rate NPV}}{(\text{Lower rate NPV} - \text{Higher Rate NPV})} \times (\text{High Rate} - \text{low Rate})$$

13. Income Statement

Particulars	Amount (₹)
Sales	-
Less: Variable cost	-
Contribution	-
Less: Fixed cost	-
Earnings Before Interest & Tax (EBIT)	-
Less: Interest	-
Earnings Before Tax	-
Less: Tax	-
Earnings After Tax	-
Less: Preference Dividend	-
Earnings Available for Equity (EAFE)	-
No. of equity shares	-
Earnings Per Share (EPS)	-

14. Cost of Equity – Dividend Approach

$$K_e = \frac{D}{P_0} \times 100$$

15. Cost of Equity – Earning Approach

$$K_e = \frac{E}{P_0} \times 100$$

16. Cost of Equity – Dividend Growth Approach or Constant Growth Approach or Gordon Model

$$K_e = \frac{D_1}{P_0} + g$$

~~✗~~ ~~✗~~

17. Cost of Equity – Earning Growth Approach

$$K_e = \frac{E_1}{P_0} + g$$

✓

18. Cost of Equity – Capital Assets Pricing Model

$$K_e = R_f + (R_m - R_f)(\beta)$$

✓

19. Cost of Equity – Realized Yield Approach

$$\text{Return of one year} = \frac{\text{dividend} + \text{Capital Gain}}{\text{Investment}}$$

$$(1 + r_1) = \frac{D_1 + P_1}{P_0}$$

$$(1 + r_2) = \frac{D_2 + P_2}{P_1}$$

$$K_e = \sqrt[n]{(1 + r_1) \times (1 + r_2) \times (1 + r_3) \dots \dots \dots (1 + r_n)} - 1$$

1) $\sqrt{\quad}$ → 12 times
 2) $- 1$
 3) $\div$ (n)
 4) $+ 1$
 5) $\times =$ → 12 times

Or If year wise price data is not given than use YTM method

20. Cost of Equity terms

D = Dividend per share = E × DP Ratio

E = Earning per share

D1 = Expected dividend per share = $D_0(1 + g)$ = E1 × DP Ratio

E1 = Expected earning per share = $E_0(1 + g)$

DP Ratio = Dividend payout ratio = $\frac{DPS}{EPS} \times 100$

P0 = Net Proceeds / Market value / Face value

g = Growth rate

Option – 1 – $g = (b)(r)$

Option – 2 – $g = \sqrt[n]{\frac{D_n}{D_0}} - 1$

b = Retention Ratio = $100 - \text{Dividend Payout Ratio}$

r = Rate of Return

Rf = Risk Free Return

Rm = Market rate of return

Rm – Rf = Market risk premium

B = Beta coefficient

21. Cost of Retained Earnings

$$\frac{K_r}{P_0 = MV/FV} = \frac{K_e}{P_0 = MV/FV}$$

If there is no flotation cost i.e. NP can't be computed

② Existing Shares

$$\frac{K_r}{P_0 = MV/FV} \neq \frac{K_e}{P_0 = NP/IP}$$

If there is flotation cost i.e. NP can be computed

② New Share Issue

In case if personal tax is given then

$$K_r = K_e(1 - t_p)(1 - B)$$

Question – 4

From the following information, calculate the cost of equity according to (a) Dividend price approach; (b) Dividend price plus growth approach; (c) Earning Price Ratio approach; (d) Earning price plus growth approach; (e) Capital assets pricing model;

- 1) Current market price of an equity share : ₹ 100 (MP)
- 2) Expected earnings per share at the end of the year : ₹ 10 (EPS₁)
- 3) Dividend Payout ratio : 80% → $D_1 = 10 \times 80\% = 8$
- 4) Growth Rate : 6% (g)
- 5) Rate of return on risk free investment : 8% (R_f)
- 6) Rate of return on market portfolio : 18% (R_m)
- 7) Volatility of securities return relative to the return of a broad based market portfolio : 1.275 (β)

$$(a) K_e = \frac{D_1}{NP} = \frac{8}{100} = 8\%$$

$$(b) K_e = \frac{D_1}{NP} + g = \frac{8}{100} + 0.06 = 14\%$$

$$(c) K_e = \frac{EPS_1}{NP} = \frac{10}{100} = 10\%$$

$$(d) K_e = \frac{EPS_1}{NP} + g = \frac{10}{100} + 0.06 = 16\%$$

$$(e) K_e = 8 + (18 - 8)(1.275) = 20.75\%$$

Solution**(a) Dividend Price Approach**

$$K_e = \frac{D_1}{P_0} = \frac{80\% \text{ of } 10}{100} = 8\%$$

(b) Dividend Price Plus Growth Approach

$$K_e = \frac{D_1}{P_0} + g = \frac{80\% \text{ of } 10}{100} + 6\% = 14\%$$

(c) Earning Price Approach

$$K_e = \frac{E_1}{P_0} = \frac{10}{100} = 10\%$$

(d) Earning Price plus Growth Approach

$$K_e = \frac{E_1}{P_0} + g = \frac{10}{100} + 6\% = 16\%$$

(e) Capital Assets Pricing Model

$$K_e = R_f + \beta \times (R_m - R_f) = 8\% + 1.275 \times (18\% - 8\%) = 20.75\%$$

Question – 5

The shares of a chemical company are selling at ₹ 20 per share. The firm had paid dividend @ ₹ 2 per share last year. The estimated growth of the company is approximately 5% per year.

- (a) Determine the cost of equity capital of the company. $\rightarrow \frac{2(1+0.05)}{20} + 0.05 = 15.5\%$
- (b) Determine the estimated market price of the equity share if the anticipated growth rate of the firm
- (i) rises to 8% $\rightarrow K_e = \frac{D_1}{NP} + g \Rightarrow 0.155 = \frac{2(1+0.08)}{NP} + 0.08 \Rightarrow NP =$
- (ii) falls to 3% $\rightarrow K_e = \frac{D_1}{NP} + g \Rightarrow 0.155 = \frac{2(1+0.03)}{NP} + 0.03 \Rightarrow NP =$

Solution

- (a) Net Proceeds (P0) = ₹ 20

Next expected dividend (D1) = $D_0 \times (1+g) = 2 \times (1+0.05) = ₹ 2.10$

$$\text{Cost of equity (Ke)} = \frac{D_1}{P_0} + g = \frac{2.10}{20} + 0.05 = 0.155 = 15.50\%$$

- (b) (i) Growth rate (g) = 8% = 0.08

$$K_e = \frac{D_1}{P_0} + g$$

$$0.155 = \frac{2(1+0.08)}{P_0} + 0.08$$

$$0.075 = \frac{2.16}{P_0}$$

$$P_0 = ₹ 28.80$$

- (ii) Growth rate (g) = 3% = 0.03

$$K_e = \frac{D_1}{P_0} + g$$

$$0.155 = \frac{2(1+0.03)}{P_0} + 0.03$$

$$0.125 = \frac{2.06}{P_0}$$

$$P_0 = ₹ 16.48$$

Question – 6

Following data relates to SK Ltd.:

Year	1	2	3	4	5
Dividend per share at end $\rightarrow$	D ₁ 2.00	D ₂ 2.00	D ₃ 2.40	D ₄ 2.50	D ₅ 2.30
Price per share (at the beginning) $\rightarrow$	P ₀ 18.00	P ₁ 19.50	P ₂ 23.00	P ₃ 22.00	P ₄ 21.20

Calculate the cost of equity using realized yield approach.

$$1+r_1 = \frac{2+19.50}{18}$$

$$1+r_2 = \frac{2+23}{19.50}$$

$$1+r_3 = \frac{2.40+22}{23}$$

$$1+r_4 = \frac{2.50+21.20}{22}$$

Solution

Firstly, we have to compute the annual yield or return generated by the share over the years. For this purpose, we will assume that share was purchased at the beginning of year 1 (as no purchase data is provided).

$$1+Y_1 = \frac{D_1+P_1}{P_0} = \frac{2+19.50}{18.00} = 1.1944$$

$$1+Y_2 = \frac{D_2+P_2}{P_1} = \frac{2+23.00}{19.50} = 1.2821$$

$$1+Y_3 = \frac{D_3+P_3}{P_2} = \frac{2.40+22.00}{23.00} = 1.0609$$

$$1+Y_4 = \frac{D_4+P_4}{P_3} = \frac{2.50+21.20}{22.00} = 1.0772$$

$$\text{Approx. } K_e = \frac{(1000 \times 10\%) + \frac{(1128 - 1000)}{5}}{\frac{(1000 + 1128)}{2}} =$$

Handwritten notes for the above formula:

- Yr: 0, 1-5, 5
- CF: (1000), 100, 1128
- PVIF: AF, PVF, NPV
- PV: 1, 1, 1
- PVIF: 1, 1, 1
- NPV: 1, 1, 1

Now we will calculate geometric mean of the above returns to calculate the cost of equity (realized return p.a.)

$$K_e = [(1+Y_1) \times (1+Y_2) \times \dots \times (1+Y_n)]^{(1/n)} - 1$$

$$K_e = [1.1944 \times 1.2821 \times 1.0609 \times 1.0772]^{(1/4)} - 1 = 1.15 - 1 = 0.15 = 15\%$$

Question – 7

Mr. S had purchased a share of SK Limited for ₹1,000. He received dividend for a period of five years at the rate of 10%. At the end of the fifth year, he sold the share for ₹1,128. You are required to compute the cost of equity as per realised yield approach.

Solution

$$\text{Approx. } K_e = \frac{\text{Dividend} + \text{Capital Gain}}{\text{Investment}} = \frac{100 + [(1128 - 1000) \div 5]}{1000} = 12.56\%$$

Year	Cash Flows	Present Value		Present Value	
		PVF @ 12%	PV	PVF @ 13%	PV
0	1000	1.000	(1000)	1.000	(1000)
1-5	100	3.605	36.05	3.517	35.17
5	1128	0.567	639.58	0.543	612.50
NPV			0.08		(35.80)

$$\text{Cost of Debt} = L + \left[\frac{NPV_L}{NPV_L - NPV_H} \right] (H - L)$$

Eq. (K_e)

$$= 10 + \left[\frac{0.08}{0.08 - (-35.80)} \right] (13 - 12) = 12.002\%$$

$$r = \frac{1}{P/E} = \frac{1}{6.25} = 0.16 = 16\%$$

$$b = 100 - 60\% = 40\%$$

$$P/E \text{ Ratio} = \frac{MPS}{EPS}$$

$$MPS = EPS \times P/E = 20 \times 6.25 = 125$$

Question – 8

$$g = 16\% \times 0.4 = 6.4\%$$

$$D_0 = 20 \times 60\% = 12$$

JC Ltd. is planning an equity issue in current year. It has an earning per share (EPS) of ₹20 and proposes to pay 60% dividend at the current year end. With a PE ratio 6.25, it wants to offer the issue at market price. The flotation cost is expected to be 4% of the issue price.

Required: Determine the required rate of return for equity share (cost of equity) before the issue and after the issue.

$$K_e = \frac{125 - 4\% = 120}{120} + 0.064 =$$

$$K_e = \frac{D_1}{NP} + g = \frac{12(1 + 0.064)}{125} + 0.064 =$$

Solution

$$\text{Current market price (P}_0\text{)} = EPS \times PE \text{ Ratio} = 20 \times 6.25 = ₹125$$

$$\text{Rate of return (r)} = 1 \div PE \text{ Ratio} = 1 \div 6.25 = 16\%$$

$$\text{Retention ratio (b)} = 100 - \text{Dividend payout ratio} = 100 - 60\% = 40\% = 0.40$$

$$\text{Growth rate} = b \times r = 0.40 \times 0.16 = 0.064$$

$$D_0 = EPS \times \text{Dividend payout ratio} = 20 \times 60\% = ₹12$$

$$10(1+g)^5 = 12.76$$

$$(1+g)^5 = \frac{12.76}{10} \Rightarrow g = \sqrt[5]{\frac{12.76}{10}} - 1 = 5\%$$

$$K_e = 10 + (18 \times 1.25) = 32.5$$

$$D_1 = D_0 \times (1+g) = 12 \times (1+0.064) = ₹ 12.768$$

$$\text{Proceeds from new issue of shares} = 125 - (125 \times 4\%) = ₹ 120$$

$$\frac{D_1}{P_0} + g = \frac{32.5}{100}$$

$$\frac{(12.76)(1+0.064)}{NP} + 0.05 = 0.325$$

NP = ✓

$$\text{Cost of equity before issue } (k_e) = \frac{D_1}{P_0} + g = \frac{12.768}{125} + 0.064 = 0.1661 = 16.61\%$$

$$\text{Cost of equity after issue } (k_e) = \frac{D_1}{P_0} + g = \frac{12.768}{120} + 0.064 = 0.1704 = 17.04\%$$

Question – 9

A company issues:

- 15% convertible debentures of ₹ 100 each at par with a maturity period of 6 years. On maturity, each debenture will be converted into 2 equity shares of the company. The risk-free rate of return is 10%, market risk premium is 18% and beta of the company is 1.25. The company has paid dividend of ₹ 12.76 per share. Five years ago, it paid dividend of ₹ 10 per share. Flotation cost is 5% of issue amount.
- 5% preference shares of ₹ 100 each at premium of 10%. These shares are redeemable after 10 years at par. Flotation cost is 6% of issue amount.

Assuming corporate tax rate is 40%.

- Calculate the cost of convertible debentures using the approximation method
- Use YTM method to calculate the cost of preference shares.

$$\begin{array}{l} \text{CF} = 15 \\ \text{PV} = \frac{15}{1.10^5} = 8.53 \\ \text{PV of } 100 = \frac{100}{1.10^5} = 68.31 \\ \text{PV of } 200 = \frac{200}{1.10^5} = 126.61 \\ \text{PV of } 100 = \frac{100}{1.10^5} = 68.31 \\ \text{PV of } 200 = \frac{200}{1.10^5} = 126.61 \\ \text{PV of } 100 = \frac{100}{1.10^5} = 68.31 \\ \text{PV of } 200 = \frac{200}{1.10^5} = 126.61 \\ \text{PV of } 100 = \frac{100}{1.10^5} = 68.31 \\ \text{PV of } 200 = \frac{200}{1.10^5} = 126.61 \end{array}$$

Year	1	2	3	4	5	6	7	8	9	10
PVIF _{0.03,t}	0.971	0.943	0.915	0.888	0.863	0.837	0.813	0.789	0.766	0.744
PVIF _{0.05,t}	0.952	0.907	0.864	0.823	0.784	0.746	0.711	0.677	0.645	0.614
PVIFA _{0.03,t}	0.971	1.913	2.829	3.717	4.580	5.417	6.230	7.020	7.786	8.530
PVIFA _{0.05,t}	0.952	1.859	2.723	3.546	4.329	5.076	5.786	6.463	7.108	7.722

Interest rate	1%	2%	3%	4%	5%	6%	7%	8%	9%
FVIF _{i,5}	1.051	1.104	1.159	1.217	1.276	1.338	1.403	1.469	1.539
FVIF _{i,6}	1.062	1.126	1.194	1.265	1.340	1.419	1.501	1.587	1.677
FVIF _{i,7}	1.072	1.149	1.230	1.316	1.407	1.504	1.606	1.714	1.828

Solution

- As per CAPM, $K_e = R_f + [\beta \times (R_m - R_f)] = 10 + (18 \times 1.25) = 32.5\%$

Also, let growth rate = g

Now, $10(1+g)^5 = 12.76$

$$(1+g)^5 = 1.276$$

From the Interest rate table, we can say that $g = 5\%$ as for five years at 5% value is 1.276.As per Constant growth model, $K_e = \frac{D_1}{P_0} + g$

$$0.325 = \frac{12.76(1+0.05)}{P_0} + 0.05$$

$$0.275 = \frac{13.398}{P_0}$$

$$P_0 = 48.72$$

Thus, share price today = ₹ 48.72

$$K_d = \frac{15(1-0.40) + \frac{120.57-95}{6}}{95 + \frac{130.57}{2}} = 15(1-0.40) + \frac{120.57-95}{6}$$

$$PV = 2 \times 48.72 \times (1+0.05)^6 = 2 \times 48.72 \times 1.340 = 130.57$$

Redemption value will be higher of:

(a) Cash value of debenture = ₹ 100

(b) Value of equity shares = $2 \times 48.72 \times (1 + 0.05)^6 = 2 \times 48.72 \times 1.340 = ₹ 130.57$

Thus, redemption value will be ₹ 130.57

As per approximation method,

$$K_d = \frac{I(1-t) + [(RV-NP) \div n]}{[(NP+RV) \div 2]}$$

$$I = 15\% \times 100 = 15$$

$$t = 0.40$$

$$RV = 130.57 \quad NP = 100 - 5\% = 95$$

$$K_d = \frac{15(1-0.40) + [(130.57-95) \div 6]}{[(95+130.57) \div 2]} = \frac{14.93}{112.785} = 0.1324 = 13.24\%$$

(ii) Cost of Preference Shares using YTM Method:

$$\text{Preference dividend} = 5\% \times 100 = 5$$

$$\text{Redemption value} = 100 \quad \text{years to maturity} = 10$$

$$\text{Investment} = 100 + (100 \times 10\%) - (110 \times 6\%) = ₹ 103.40$$

$$\text{NPV at 5\%} = \text{PVC I} - \text{PVC O}$$

$$= \text{PV of Preference dividend} + \text{PV of Redemption Value} - \text{Investment}$$

$$= [5 \times 7.722] + [100 \times 0.614] - 103.40 = ₹ 3.39$$

$$\text{NPV at 3\%} = \text{PVC I} - \text{PVC O}$$

$$= \text{PV of Preference dividend} + \text{PV of Redemption Value} - \text{Investment}$$

$$= [5 \times 8.530] + [100 \times 0.744] - 103.40 = ₹ 13.65$$

$$\text{Cost of Preference (Kp)} = L + \left[\frac{NPV_L}{NPV_L - NPV_H} \right] (H - L) = 3 + \left[\frac{13.65}{13.65 - (-3.39)} \right] (5 - 3) = 4.60\%$$

22. Weighted Average Cost of Capital (WACC = Ko)

- It is the weighted average of cost of all sources taken together.

$$K_o = (K_e)(W_e) + (K_r)(W_r) + (K_p)(W_p) + (K_d)(W_d)$$

- Weights can be either book value, market value or target value.

- Prefer to use MV weights if question is silent, provided MV of all sources can be computed

Source	(A) Amount $\rightarrow$ BV	(B) Weights	Cost of capital	(A x B) Total cost
Equity share capital	- x	- $x \div \text{Total}$	-	-
Reserve & Surplus	- y	- $y \div \text{Total}$	-	-
Preference share capital	- z	-	-	-
Debentures	- s	-	-	-
	(-) Total	(1)	-	-

Always '1'

$$WACC = \frac{TC}{\text{Amo.}} \times 100$$

23. Points to Remember (PTRs)

- Flotation cost are not to be considered for calculating market value weights.

- Term loan doesn't have any market value. If market value is required than consider its book value to be its market value.

- We always require ex-dividend or ex-interest values.

- Ex-dividend value = Cum-dividend value – Dividend amount
- Ex-interest value = Cum-interest value – Interest amount
- Market value of an equity share represents value towards face value and reserve & surplus.
- If $K_r \neq K_e$ then distribute the total market value between face value and reserve and surplus in the ratio of their book value.

$$10(1+g)^7 = 14.07$$

$$(1+g)^7 = 1.407$$

$$g = 5\%$$

$$K_e = \frac{16}{80} + 0.05 = 25\%$$

$$K_p = 8 + \frac{(106-104)}{\frac{104+106}{2}} = 8\%$$

Question – 10

Capital structure of D Ltd. as on 31st March, 2023 is given below:

Particulars	TC	Cost %	₹
Equity share capital (₹ 10 each)	7.58	25%	30,00,000
8% Preference share capital (₹ 100 each)	0.80	8%	10,00,000
12% Debentures (₹ 100 each)	0.90	9.01%	10,00,000
	9.28		500

$$K_d = 12(1-0.40) + \frac{(120-95)}{\frac{195+120}{2}} = 9.01\%$$

$$K_o = \frac{9.28}{500} \times 100 = 18.40\%$$

- Current market price of equity share is ₹ 80 per share. The company has paid dividend of ₹ 14.07 per share. Seven years ago, it paid dividend of ₹ 10 per share. Expected dividend is ₹ 16 per share.
- 8% Preference shares are redeemable at 6% premium after five years. Current market price per preference share is ₹ 104.
- 12% debentures are redeemable at 20% premium after 10 years. Flotation cost is ₹ 5 per debenture.
- The company is in 40% tax bracket.
- In order to finance an expansion plan, the company intends to borrow 15% Long-term loan of ₹ 30,00,000 from bank. This financial decision is expected to increase dividend on equity share from ₹ 16 per share to ₹ 18 per share. However, the market price of equity share is expected to decline from ₹ 80 to ₹ 72 per share, because investors' required rate of return is based on current market conditions.

$$K_e = \frac{18}{72} + 0.05 = 30\%$$

Required:

- Determine the existing Weighted Average Cost of Capital (WACC) taking book value weights.
- Compute Weighted Average Cost of Capital (WACC) after the expansion plan taking book value weights.

Interest Rate	1%	2%	3%	4%	5% ✓	6%	7%
FVIF _{i,5}	1.051	1.104	1.159	1.217	1.276	1.338	1.403
FVIF _{i,6}	1.062	1.126	1.194	1.265	1.340	1.419	1.501
FVIF _{i,7}	1.072	1.149	1.230	1.316	1.407	1.504	1.606

Solution

- (a) Growth rate in dividend

$$14.07 = 10 \times FVIF_{(i,7 \text{ years})}$$

$$FVIF_{(i,7 \text{ years})} = 1.407$$

$$FVIF_{(5\%, 7 \text{ years})} = 1.407$$

$$i = 5\%$$

Growth rate in dividend = 5%

- (b) $K_e = \frac{D_1}{P_0} + g = \frac{16}{80} + 0.05 = 25\%$

$$(c) K_p = \frac{PD + \left(\frac{RV-NP}{n}\right)}{\left(\frac{RV+NP}{2}\right)} = \frac{8 + \left(\frac{106-104}{5}\right)}{\left(\frac{106+104}{2}\right)} = 8\%$$

$$(d) K_d = \frac{I(1-t) + \left(\frac{RV-NP}{n}\right)}{\left(\frac{RV+NP}{2}\right)} = \frac{12(1-0.4) + \left(\frac{120-95}{10}\right)}{\left(\frac{120+95}{2}\right)} = 9.02\%$$

(i) Statement of WACC

Source	Book Value	Cost of capital	Total cost
Equity share capital	30,00,000	25%	7,50,000
Preference share capital	10,00,000	8%	80,000
Debentures	10,00,000	9.02%	90,200
	50,00,000		9,20,200

$$WACC = \frac{9,20,200}{50,00,000} \times 100 = 18.40\%$$

$$(ii) \text{ Cost of long term debt} = 15(1 - 0.40) = 9\%$$

$$\text{Revised } K_e = \frac{18}{72} + 0.05 = 30\%$$

Statement of WACC

Source	Book Value	Cost of capital	Total cost
Equity share capital	30,00,000	30%	9,00,000
Preference share capital	10,00,000	8%	80,000
Debentures	10,00,000	9.02%	90,200
Long term debt	30,00,000	9%	2,70,000
	80,00,000		13,40,200

$$WACC = \frac{13,40,200}{80,00,000} \times 100 = 16.76\%$$

Question – 11

The latest Balance Sheet of SK Ltd. is given below:

(₹ '000)

Ordinary shares (50,000 shares)	→ 500
✓ Share Premium	→ 100
✓ Retained profits	→ 600
	Eq. Sh. fund = 1,200
8% Preference shares	→ 400
13% Perpetual debts (Face value ₹ 100 each)	→ 600
	→ 2,200

The ordinary shares are currently priced at ₹ 39 ex-dividend each and ₹ 25 preference share is priced at ₹ 18 cum-dividend. The debentures are selling at 110% ex-interest and tax is paid by SK Ltd. at 40%. SK Ltd. has a beta of 0.90, risk free return is 10% & market return is 20%. Calculate the weighted average cost of capital, (based on market value) WACC of SK Ltd.

$$K_e = 10 + (20 - 10)(0.90) = 19\%$$

$$K_s = K_e = 19\%$$

$$K_p = \frac{2}{16} = 12.50\%$$

$$K_d = \frac{13(1-0.40)}{110} = 7.09\%$$

Solution

$$\text{Cost of equity (K}_e\text{)} = R_f + (R_m - R_f)(\beta) = 10 + (20 - 10)(0.90) = 19\%$$

$$\text{Since there is no flotation cost, thus cost of retained earning (K}_r\text{)} = K_e = 19\%$$

$$\text{Price of preference share ex-dividend} = 18 - (25 \times 8\%) = 18 - 2 = ₹ 16$$

$$\text{Cost of preference shares (K}_p\text{)} = \frac{\text{Pref. Dividend}}{P_0} = \frac{25 \times 8\%}{16} = 12.5\%$$

$$\text{Market price of debenture} = 100 \times 110\% = ₹ 110$$

$$\text{Cost of debt (K}_d\text{)} = \frac{I(1-t)}{P_0} = \frac{100 \times 13\%}{110} \times 100 = 7.09\%$$

Calculation of weighted average cost of capital

Source	Market Value (₹) (A)	Cost (B)	A × B
Equity shareholder fund	$50,000 \times 39 = 19,50,000$	19%	3,70,500
Preference Share	$\frac{4,00,000}{25} \times 16 = 2,56,000$	12.50%	32,000
Debentures	$\frac{6,00,000}{100} \times 110 = 6,60,000$	7.09%	46,794
	28,66,000		4,49,294

$$\text{Weighted average cost of capital} = \frac{4,49,294}{28,60,000} \times 100 = 15.68\%$$

Question – 12

Determine the cost of capital of SK Ltd. using the book value and market value weights from the following information:

Sources	Book Value (₹)	Market Value (₹)
Equity shares	1,20,00,000	2,00,00,000
Retained earnings	30,00,000	-
Preference shares	36,00,000	33,75,000
Debentures	9,00,000	10,40,000

Additional information:

$$K_r = \frac{15}{130} + g$$

$$K_e = \frac{15}{120} + g$$

(a) Equity: Equity shares are quoted at ₹ 130 per share and a new issue priced at ₹ 125 per share will be fully subscribed; flotation cost will be ₹ 5 per share.

(b) Dividend: During the previous 5 years, dividends have steadily increased from ₹ 10.60 to ₹ 14.19 per share. Dividend at the end of the current year is expected to be ₹ 15 per share.

(c) Preference shares: 15% Preference shares with face value of ₹ 100 would realise ₹ 105 per share.

(d) Debentures: The company proposes to issue 11-year 15% debentures but the yield on debentures of similar maturity and risk class is 16%; flotation cost is 2%.

(e) Tax: Corporate tax rate is 35%. Ignore dividend tax.

(f) Flotation cost would be calculated on face value.

$$NP = 93.75 - (2\% \times 100)$$

$$= 91.75$$

$$10.6(1+g)^5 = 14.19$$

$$g = \checkmark$$

$$K_p = \frac{15}{105} = \checkmark$$

$$IP = \frac{15\% \times 100}{16\%} = 93.75$$

Solution

- (i) Calculation of
- g
- :

$$₹10.6(1 + g)^5 = ₹14.19$$

$$\text{Or, } (1 + g)^5 = \frac{14.19}{10.6} = 1.338$$

Table (FVIF) suggests that ₹ 1 compounds to ₹ 1.338 in 5 years at the compound rate of 6 percent. Therefore, g is 6 per cent.

$$\text{Cost of Equity } (K_e) = \frac{D_1}{P_0 - F} + g = \frac{₹15}{₹125 - ₹5} + 0.06^* = 0.125 + 0.06 = 0.185$$

$$(ii) \text{ Cost of Retained Earnings } (K_r) = \frac{D_1}{P_0} + g = \frac{₹15}{₹125} + 0.06 = 0.18$$

$$(iii) \text{ Cost of Preference Shares } (K_p) = \frac{PD}{P_0} = \frac{₹15}{₹105} = 0.1429$$

$$(iv) \text{ Cost of Debentures } (K_d) = \frac{I(1-t) + \left(\frac{RV - NP}{n}\right)}{\frac{RV + NP}{2}} = \frac{₹15(1-0.35) + \left(\frac{₹100 - ₹91.75^*}{11 \text{ years}}\right)}{\frac{₹100 + ₹91.75^*}{2}} = 0.1095$$

*Since yield on similar type of debentures is 16 per cent, the company would be required to offer debentures at discount.

$$\text{Market price of debentures (approximation method)} = ₹15 \div 0.16 = ₹93.75 \rightarrow \text{IP}$$

$$\text{Sale proceeds from debentures} = ₹93.75 - ₹2 \text{ (i.e., flotation cost)} = ₹91.75$$

OR Market value (P_0) of debentures can also be found out using the present value method:

$$P_0 = \text{Annual Interest} \times \text{PVIFA } (16\%, 11) + \text{Redemption value} \times \text{PVIF } (16\%, 11)$$

$$P_0 = ₹15 \times 5.029 + ₹100 \times 0.195 = ₹94.935$$

$$\text{Net Proceeds} = ₹94.935 - 2\% \text{ of } ₹100 = ₹92.935$$

Accordingly, the cost of debt can be calculated

Total Cost of capital [BV weights and MV weights] (Amount in (₹) lakh)

Source of Capital	Weights		Specific cost (K)	Total Cost	
	BV	MV		(BV × K)	(MV × K)
Equity shares	120	160*	0.1850	22.2	29.6
Retained Earnings	30	40*	0.1800	5.4	7.2
Preference Shares	36	33.75	0.1429	5.14	4.82
Debentures	9	10.4	0.1095	0.986	1.139
	195	244.15		33.73	42.76

*Market Value of equity has been apportioned in the ratio of Book Value of equity and retained earnings i.e., 120:30 or 4:1.

Weighted Average Cost of Capital (WACC):

$$\text{Using Book Value} = \frac{₹33.73}{₹195} = 0.1729 \text{ or } 17.29\%$$

$$\text{Using Market Value} = \frac{₹42.76}{₹244.15} = 0.1751 \text{ or } 17.51\%$$

Question – 13

Capital Structure of B Ltd. for the year ended 31st March, 2025 are as follows:

Particulars	Amount (₹)
Equity share capital @₹ 10 each	14,00,000
10% Preference share capital @₹ 1,000 each	10,00,000
Debenture @₹ 100 each	9,60,000
Bank loan	6,40,000

- Risk-free rate of return is 14%, Market rate of return is 19% and beta of company is 1.20.
- 10% Preference shares are redeemable at ₹ 1,065.40 after 3 years.
- Interest on bank loan is 1.30 times of interest on debentures.
- Debentures are redeemable at par after 5 years. Floatation cost is ₹ 4 per debenture.
- Tax rate is 30%.
- Cost of capital is 14%

Handwritten calculations:

$$K_e = 14 + (19 - 14)(1.20) = 20\%$$

$$\text{Deb. Int.} = y \quad \therefore BL = 1.3y$$

$$K_d = \frac{(100y)(1-0.30) + \left(\frac{100-96}{5}\right)}{\left(\frac{96+100}{2}\right)} \quad K_d = 1.3y(1-0.3) = 0.91y$$

$$= \frac{70y + 0.8}{98}$$

You are required to calculate the following:

- Cost of Equity.
- Cost of preference share using YTM method.
- Post-tax cost of debenture using approximation method.
- Interest rate of bank loan.

PV factors @ 10% and 14%

Year	1	2	3	4
PVIF _{0.10,t}	0.909	0.826	0.751	0.683
PVIF _{0.14,t}	0.877	0.769	0.675	0.592

Solution

- Cost of equity = $K_e = R_f + (R_m - R_f)(B) = 14 + (19 - 14)(1.2) = 20\%$

- Cost of Preference Shares using YTM

Preference dividend = $10\% \times 1000 = 100$

Redemption value = 1,065.40 years to maturity = 3

Investment = ₹ 1000

Year	Cash Flows	PVF @10%	PV (₹)	PVF @14%	PV (₹)
0	(1,000)	1	(1,000)	1	(1,000)
1 - 3	100	2.487	248.70	2.322	232.20
3	1065.40	0.751	800.12	0.675	719.14
			+48.82		-48.66

$$\text{Cost of Preference (Kp)} = L + \left[\frac{NPV_L}{NPV_L - NPV_H} \right] (H - L) = 10 + \left[\frac{48.82}{48.82 - (-48.66)} \right] (14 - 10) = 12\%$$

- Let the rate of interest on debenture be 'y'

Thus, rate of interest on bank loan = $1.3y$

Cost of bank loan = $Kd1 = 1.3y \times (1 - 0.3) = 0.91y$

Cost of Debenture = $Kd2 = \frac{I(1-t) + \left(\frac{RV-NP}{n}\right)}{\frac{RV+NP}{2}} = \frac{(100y)(1-0.30) + \left(\frac{₹100-₹96}{5 \text{ years}}\right)}{\frac{₹96+₹100}{2}} = \frac{70y+0.8}{98}$

Weighted Average Cost of Capital

Source of Capital	Amount	Weight (W)	Cost (C)	W × C
Equity share capital	14,00,000	0.35	0.20	0.07
Preference Shares	10,00,000	0.25	0.12	0.03
Debentures	9,60,000	0.24	$\frac{70y+0.8}{98}$	$\frac{16.8y+0.192}{98}$
Bank Loan	6,40,000	0.16	$0.91y$	$0.1456y$
	40,00,000	1		0.14

$$0.14 = 0.07 + 0.03 + \frac{16.8y+0.192}{98} + 0.1456y$$

$$(0.14 - 0.07 - 0.03) \times 98 = 16.8y + 0.192 + (0.1456y)(98)$$

$$3.92 = 16.8y + 0.192 + 14.2688y$$

$$3.728 = 31.0688y$$

$$y = 0.1199 \text{ or } 12\%$$

So interest on debenture = 12%

$$\text{Cost of debenture} = \frac{70(12\%)+0.8}{98} = \frac{9.2}{98} = 9.39\%$$

(iv) Interest rate of bank loan = $1.3y = 1.3 \times 12 = 15.60\%$

24. Weighted Marginal Cost of Capital

It is the cost of raising additional rupee of capital.

In this case only the additional capital raised should be considered for calculating K_o .

Source	Amount	Weights	Cost of capital	Total cost
Equity share capital	-	-	-	-
Reserve & Surplus	-	-	-	-
Preference share capital	-	-	-	-
Debentures	-	-	-	-
	-	-	-	-

Additional

Question – 14

Following are the information of TT Ltd.:

Particulars	
Earnings per share	→ ₹ 10
Dividend per share	→ ✓ ₹ 6
Expected growth rate in Dividend (a)	→ 6%
Current market price per share MP	→ ₹ 120
Tax rate	→ 30%
Requirement of Additional Finance	→ ₹ 30 lakhs
Debt Equity Ratio (For additional finance)	→ (2:1)
Cost of Debt	
0 - 5,00,000	10%
5,00,001 – 10,00,000	✓ 9%
Above 10,00,000	✓ 8%

Handwritten calculations for Cost of Debt:

$$\text{Debt} = 30 \times \frac{2}{3} = 20$$

$$\text{Eq.} = 30 \times \frac{1}{3} = 10$$

Weighted average cost of debt after tax:

$$\frac{(5 \times 10\%) + (5 \times 9\%) + (10 \times 8\%)(1 - 0.30)}{20} = 6.125\%$$

Assuming that there is no Reserve and Surplus available in TT Ltd. You are required to:

- Find the pattern of finance for additional requirement
- Calculate post tax average cost of additional debt
- Calculate cost of equity → $\frac{6(1+0.06)}{120} + 0.06 = 11.30\%$
- Calculate the overall weighted average after tax cost of additional finance

Solution

(a) Pattern of raising capital

Debt ($30,00,000 \times 2/3$) = ₹ 20,00,000

Equity ($30,00,000 \times 1/3$) = ₹ 10,00,000

Equity Fund:

Equity (additional) = ₹ 10,00,000

₹ 10,00,000

Debt Fund:

10% Debt = ₹ 5,00,000

9% Debt = ₹ 5,00,000

8% Debt = ₹ 10,00,000

₹ 20,00,000

(b) $K_d = \frac{\text{Interest} (1-t)}{P_0} \times 100 = \frac{[(5,00,000 \times 10\%) + (5,00,000 \times 9\%) + (10,00,000 \times 8\%)](1-0.30)}{20,00,000} \times 100$

$$= \frac{1,22,500}{20,00,000} \times 100 = 6.125\%$$

(c) $K_e = \frac{D(1+g)}{P_0} + g = \frac{6 \times (1+0.06)}{120} + 0.06 = \frac{5.36}{120} + 0.06 = 0.113 = 11.3\%$

(d) Weighted average cost of capital

Source	Amount (₹)	Weight	Cost of capital after tax	WACC
Equity Fund	✓ 10,00,000	1/3	→ 11.3	3.767
Debt Fund	✓ 20,00,000	2/3	→ 6.125	4.083
Total	30,00,000	1		7.85

$$2.476(1+g) = 2.773$$

Question – 15

$$g = 0.12 = 12\%$$

The SK Company has following capital structure at 31st March, 2021 which is considered to be optimum:

13% debenture ₹ 3,60,000

weights

15%

11% Preference share capital ₹ 1,20,000

5%

Equity share capital (2,00,000 shares) ₹ 19,20,000

80%

100%

$$D_1 = 2.773 \times 50\% = 1.3865$$

The company's share has a current market price of ₹ 27.75 per share. The expected dividend per share in next year is 50% of the 2021 EPS. The EPS of last 10 years is as follows. The past trends are expected to continue:

Year	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
EPS	1.000	1.120	1.254	1.405	1.574	1.762	1.974	2.211	2.476	2.773

The company can issue 14% new debenture. The company's debenture is currently selling at ₹ 98. The new preference issue can be sold at a net price of ₹ 9.80, paying a dividend of ₹ 1.20 per share. The company's marginal tax rate is 50%.

$$\frac{14(1-0.50)}{98} =$$

$$\frac{1.20}{9.80} =$$

(a) Calculate the after tax cost (i) of a new debts and new preference share capital, (ii) of ordinary equity assuming new equity comes from the retained earnings. → $K_e = \frac{1.3865}{27.75} + 0.12 =$

(b) Calculate the marginal cost of capital

(c) How much can be spent for capital investment before new ordinary share must be sold? Assuming that retained earnings available for next year's investment are 50% of 2021 earnings.

(d) What will be marginal cost of capital {cost of fund raised in excess of the amount calculated in part (c)} if the company can sell new ordinary shares to net ₹ 20 per share? The cost of debt and of preference capital is constant.

$$K_e = \frac{1.3865}{20} + 0.12 =$$

$$\begin{aligned} \text{Total Inv.} \times 80\% &= \text{Eq. Sh.} \\ \text{Total Inv.} \times 80\% &= 2.773 \times 50\% \times 20 \\ \text{Inv.} &= \frac{2.773 \times 100}{80\%} = 346.625 \end{aligned}$$

Solution**Existing Capital Structure Analysis**

Source of Capital	Amount	Ratio
Equity	19,20,000	0.80
Preference Shares	1,20,000	0.05
Debentures	3,60,000	0.15
	24,00,000	1

(a) (i) Cost of new debt = $K_d = \frac{I(1-t)}{P_0} = \frac{(14\% \times 100)(1-0.50)}{98} = 0.07143 = 7.143\%$

Cost of new preference shares = $K_p = \frac{PD}{P_0} = \frac{1.20}{9.80} = 0.12245 = 12.245\%$

$$(ii) \text{ Cost of retained earnings} = K_r = \frac{D_1}{P_0} + g = \frac{2.773 \times 50\%}{27.75} + 0.12 = 0.17 = 17\%$$

$$\text{Here } g = \frac{2.773}{2.476} - 1 = 0.12 = 12\%$$

(b) **Marginal Cost of Capital**

Source of Capital	Weight	Cost	WMCC
Equity	0.80	0.17	0.1360
Preference Shares	0.05	0.12245	0.0061
Debentures	0.15	0.07143	0.0107
	1.00		0.1528 or 15.28%

(c) Amount of retained earnings available = $2.773 \times 50\% \times 2,00,000 = ₹ 2,77,300$

The ratio of equity in the total capital is 80%.

Therefore, investment that can be done before issuing new equity shares = $\frac{2,77,300}{80\%} = ₹ 3,46,625$

(d) Cost of new issue of equity shares = $\frac{D_1}{P_0} + g = \frac{2.773 \times 50\%}{20} + 0.12 = 0.1893 = 18.93\%$

Marginal Cost of Capital

Source of Capital	Weight	Cost	WMCC
Equity	0.80	0.1893	0.1514
Preference Shares	0.05	0.12245	0.0061
Debentures	0.15	0.07143	0.0107
	1.00		0.1682 or 16.82%